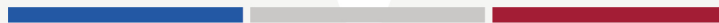




MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 3 UNIT 2
Media Set Video 1 Transcript

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Introduction

PAUL MCDONAGH-SMITH: So we're very pleased to have Professor Tim Kraska here with us today. Tim, would you like to take a moment to introduce yourself to our audience?

TIM KRASKA: Sure. So I'm Tim Kraska. I'm a professor of computer science at MIT. Until recently, I was also a director of applied science at Amazon, mainly working on agents. I'm currently also the codirector of MIT's Generative AI Impact Consortium.

Assigning autonomy

MCDONAGH-SMITH: Well, it really is a pleasure and a privilege to have you here with us. So, Tim, when I look at your research trajectory, it tells a fascinating story. You've shown that machine learning models could replace traditional database indexes, a fairly bounded optimization task, and progressively moved towards self-optimizing databases with SageDB. And now, of course, fully autonomous AI agents. Each step that you've taken expanded the scope of what you were willing to let AI systems decide without a human in the loop. For our business leaders here with us today, who are charting their own AI-adoption journeys, what did you learn at each stage about when it's safe to hand over more autonomy to an AI system, and what signals told you "not yet"?

That's an excellent question. So when we started the whole work around, like, learned indexes, machine learning for systems, our goal was mainly to make systems either easier to use or more scalable, performant, efficient. And, in the very beginning, we integrated, like, particular machine learning algorithms very deeply into the system while having exactly the same guarantees to the outside.

And so, for example, with learned indexes, we figured out on how we can use machine learning model to replace what's called a B-tree structure, which is a very old technique around since the, I would say, 80, 90s as a concept at least. And we just showed that you that we can replace this tree structure with something which is essentially a learned model. And we spent a lot of energy on trying to figure out how we can give exactly the same guarantees in the end. Meaning, like, these B-tree structures, they guarantee you that if the key exists, they will find it. If it's not there, it gives you the next closest key to it. And so we mimicked exactly that same contract just under the hood it was very differently implemented.

Many of our work on the M Alpha system set was all in that area. So, like, how do you take a model and then build infrastructure surrounding it so that the guarantees stay the same. So deploying in those cases actually becomes relatively easy in the sense of, like, if you can show it improves the average, then you know it's still correct because that's on how we designed everything.

Still, there are challenges in deployment in the sense of corner cases, and I can tell you a little bit more about that. With agents, it's quite different. You're kind of willing to take a different behavior and you know that mistakes might happen. And I think there's a lot of

the agent work like the whole approach is even different, because it's more like I would say doing what humans do which also makes mistakes, right, and now we can just have machines do similar things.

Two, three years ago, the models were simply not at the place that was even working. We didn't even consider it. Now that changed quite rapidly. So many things which I thought wouldn't be possible, like, two, three years ago, they are suddenly, they work extremely well. And so we are able to accept more of, like, this stochastic behavior because in the end, we are replacing our processes which normally humans do, again, and they also make mistakes. So I'm less worried about it.

MCDONAGH-SMITH: It fundamentally means kind of a recalibration of how we're thinking about humans and machines working together.

KRASKA: Correct.

MCDONAGH-SMITH: So Tim, you've been refreshingly candid in describing generative AI as overhyped, but real. And your KramaBench research puts hard numbers behind that claim, showing that single-agent approaches achieved only around 22% accuracy on real-world data science tasks. However, at the same time, you built production AI agents at AWS that serve thousands, tens of thousands of employees. How should a business leader hold these two truths simultaneously, that the technology is genuinely transformative and that it fails more often than vendor pitches might suggest? What does honest strategic planning look like in that space?

KRASKA: That's a really good question, man. I wish that I have a playbook for it because it's like I myself still struggle to sometimes figure out, okay, what's like really cool demos and what's actually really working. I sometimes joke that the biggest problem right now in industry are the intern demos because they are like these, models make it so easy to create a very impressive demo in a short amount of time and you see the demo and you say like, "Oh my God, we need to immediately deploy it." But then, in the moment you try to deploy it, figure out all these corner cases which don't really work yet. And the actual usefulness is quite limited in the end at least if you put it at scale.

My recommendation right now, probably for most companies, is start small but try a lot.

MCDONAGH-SMITH: Yep.

KRASKA: And so, like, I think we see tremendous benefits in more restrictive domains in the end. I use, for example, coding tools almost on a daily basis by now, and I think these tools are unbelievably good, and we have strong opinions on how they should look like. I'm a big fan by now, so I have my little plugin I use for writing every single email I do using AI. So I rarely if you get an email from me, it's rarely written by me alone anymore. I just created this very specific for me and I have this little button which uses my agent actually to do the writing. I don't like the standard out of Gmail or the Copilot from Microsoft. They don't have my voice, so I created my own little thing which helps me a lot.

Or I have this little project where I my goal is that I automate all boring tasks I have to do. So I started with reimbursements. Now I have a script which I can use through an agent system, which automatically collects all the receipts in my email inbox, right, like the Lyft receipts, Uber, and so on. It goes through, takes the flight data, it puts everything into a

certain folder structure with dates on it and why I did the trip from my calendar, and then everything gets packaged and sent to my system for the reimbursement, because this process of just finding all the emails and converting them to PDFs it's not a lot of work, but it takes time. And so I want, my dream is I drive into work and I can just say like, oh yeah, do these things.

MCDONAGH-SMITH: And just work.

KRASKA: And just work. Yeah. So what I would recommend particular big companies like give your workforce these tools, make them available, and see what sticks, and really provides value by let them explore in the end. And you can have strategic initiatives in certain areas, but like overall, like almost a bottom-up approach can help a lot. The really large initiatives around like we need to build the internal company agents which is able to do everything, like my feeling is right now they often fail more than they succeed. There are some exceptions, but, many of them I hear about, they are failing.

There are other things you can do to help the whole ecosystem. There are steps you can take. For example, building an ontology helps a lot. There's something around coding which you can enable a lot of people with. But, like, overall, I would do a bottom-up approach.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.